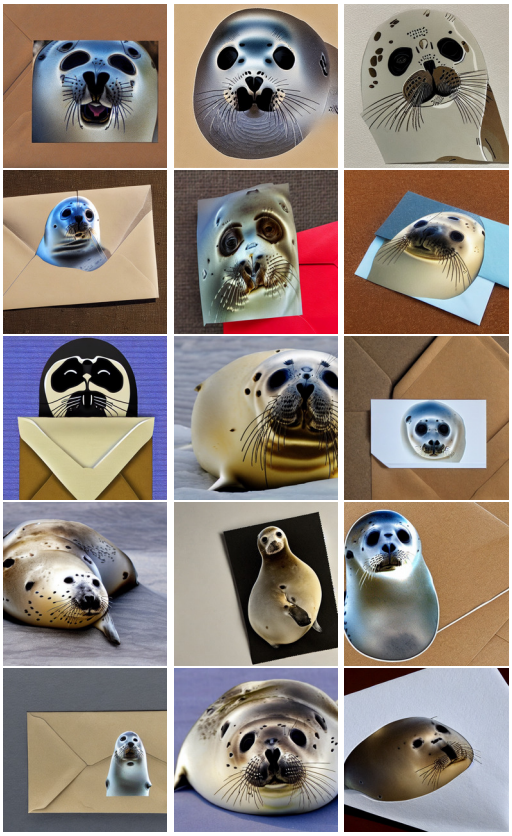
(a) Unedited prompt encoding

(b) Encoding edited to favour sense of wax or official seal



(c) Encoding edited to favour animal sense

Figure 22: Prompt: "*a seal on an envelope*"

(a) Unedited prompt encoding

(b) Encoding edited to favour luggage-related sense

(c) Encoding edited to favour tree-related sense

Figure 23: Prompt: "*a trunk*"

B Statistical Comparisons

In this section we provide detailed information about the content of the images generated in our experiments, and provide details on statistical significance testing.

B.1 Summing Encodings

The information in this section concerns the experiments described in §3.

For each pair of prompts, $\mathbf{s}_1$ and $\mathbf{s}_2$, 30 images were generated from each prompt, and 30 were generated from their weighted sum.

We were interested in the frequency with which both $\mathbf{s}_1$ and $\mathbf{s}_2$ were realised in an image generated by the weighted sum of their encodings. To ensure that this was not a coincidence, statistical significance was established using an unpaired permutation test where the statistic concerned was the difference between the number of images in which both prompts were realised in the images generated from the weighted sum and the images generated from each prompt alone. Some prompt pairs that we initially tried were discarded because they co-occurred too frequently in images generated by one of the prompts (for example, $\mathbf{s}_1$ =“*a dog in a public square*” and $\mathbf{s}_2$ =“*a tree in a public square*” were discarded because trees appear frequently in images generated by $\mathbf{s}_1$ alone, and thus a large number of samples would have been required to establish a statistically significant difference).

All weighted sums in this section were found to realise both $\mathbf{s}_1$ and $\mathbf{s}_2$ more frequently than either prompt alone to a statistically significant extent ($p < 0.05$).

	Number containing:			
Input	Dog	Lake	Both	Neither
CLIP(“*dog*”)	30	0	0	0
CLIP(“*lake*”)	0	30	0	0
$0.5 \cdot$ (CLIP(“*dog*”)) $+ 0.5 \cdot$ (CLIP(“*lake*”))	3	11	**16**	0

Table 3: Frequency of realisation of each concept for $\mathbf{s}_1$ =“*dog*”, $\mathbf{s}_2$ =“*lake*”

	Number containing:			
Input	Cat	Tree	Both	Neither
CLIP(“*cat*”)	30	0	0	0
CLIP(“*tree*”)	0	30	0	0
$0.5 \cdot$ (CLIP(“*cat*”)) $+ 0.5 \cdot$ (CLIP(“*tree*”))	6	2	**22**	0

Table 4: Frequency of realisation of each concept for $\mathbf{s}_1$ =“*cat*”, $\mathbf{s}_2$ =“*tree*”

	Number containing:			
Input	Bear	Hat	Both	Neither
CLIP(“*bear*”)	30	0	0	0
CLIP(“*hat*”)	0	30	0	0
$0.5 \cdot$ (CLIP(“*bear*”)) $+ 0.5 \cdot$ (CLIP(“*hat*”))	20	3	**5**	2

Table 5: Frequency of realisation of each concept for $\mathbf{s}_1$ =“*bear*”, $\mathbf{s}_2$ =“*hat*”